

Carlos Jurado

🌐 Academic Website — ✉ carlosjurado@utexas.edu

Education

PhD, Astronomy; The University of Texas at Austin Aug. 2024 – June 2029
BS, Astrophysics; University of California, Los Angeles (UCLA) Oct. 2020 – June 2024

Research

Graduate Student Researcher at UT Austin – Austin, TX Sept. 2024 – Present
Advisor: Keith Hawkins

Student Researcher at University of Notre Dame – Notre Dame, IN May 2023 – Aug. 2024
Advisor: Lauren Weiss

Combined radial velocity measurements of Kepler-34 and Kepler-35 from several spectrographs to place a strong mass upper limit on additional stellar and substellar companions in the two systems.

Student Researcher at UCLA – Los Angeles, CA May 2022 – June 2024
Advisor: Smadar Naoz

Utilized supercomputers to create population synthesis models and explore the spatial distribution of stellar remnants at the Galactic Center.

Publications

[1] **Jurado, C.**; Hawkins, K.; Hunt J. A. S.; Hackshaw, Z.; Fillion C.; Frankel, N., Carr, C., “Dynamical Origins of Azimuthal Metallicity Variations in the Galactic Disk: Insights from kinematic ridges with Gaia”, Accepted in ApJ, [arXiv:2601.02494](https://arxiv.org/abs/2601.02494)

[2] **Jurado, C.**; Weiss, L. M.; Daclison L.; Tofflemire, B.M.; Orosz J. A.; Welsh, W. F., “Upper Limits on Stellar Companions to the Kepler-34 and Kepler-35 Systems”, AJ, [10.3847/1538-3881/ada5f5](https://doi.org/10.3847/1538-3881/ada5f5)

[3] **Jurado, C.**; Naoz, S.; Lam, C. Y.; Hoang, B.-M., “Natal Kicks from the Galactic Center and Implications on their Environment and the Roman Space Telescope,” ApJ, doi.org/10.3847/1538-4357/ad55ee

Presentations

C. Jurado, S. Naoz, C. Y. Lam, and B.-M. Hoang, “The implications of Exploding Stars at the Galactic Center,” in Bulletin of the American Physical Society, APS Far West Section Oct. 7 2023

AstroBash, UCLA Department of Physics and Astronomy Oct. 18 2023

AstroBash, UCLA Department of Physics and Astronomy Nov. 7 2022

Teaching

Graduate Teaching Assistant, Search For Extraterrestrial Life (AST 309L)	Aug. 2025 – Dec. 2025
Graduate Teaching Assistant, Introductory Astronomy (AST 301)	Jan. 2025 – May 2025
Undergraduate Course Reader, Introductory Physics (Physics 5A)	Oct. 2023 – Dec. 2023
Undergraduate Course Reader, Introductory Mechanics (Physics 1A)	April 2024 – June 2024
Undergraduate Learning Assistant, Introductory Mechanics (Physics 1A)	Jan. 2023 – Mar. 2023

Observing Experience

McDonald Observatory <i>PI</i> : Carlos Jurado (6 nights)	Feb. 2026
<i>Spectroscopically observe 7 stellar triples in order to determine the level of chemical homogeneity.</i>	

Outreach

Student Panelist, UCLA AstroLive Summer Workshop	July 19 2022
Volunteer and Student Panelist, UCLA Physics and Astronomy Outreach Event	May 19 2022

Skills

Programming Languages: Python, C/C++
Tools: iSpec, SAPHIRES
Software: BACCHUS, IRAF, SAOImageDS9
Other: Linux/Unix, Git, L^AT_EX